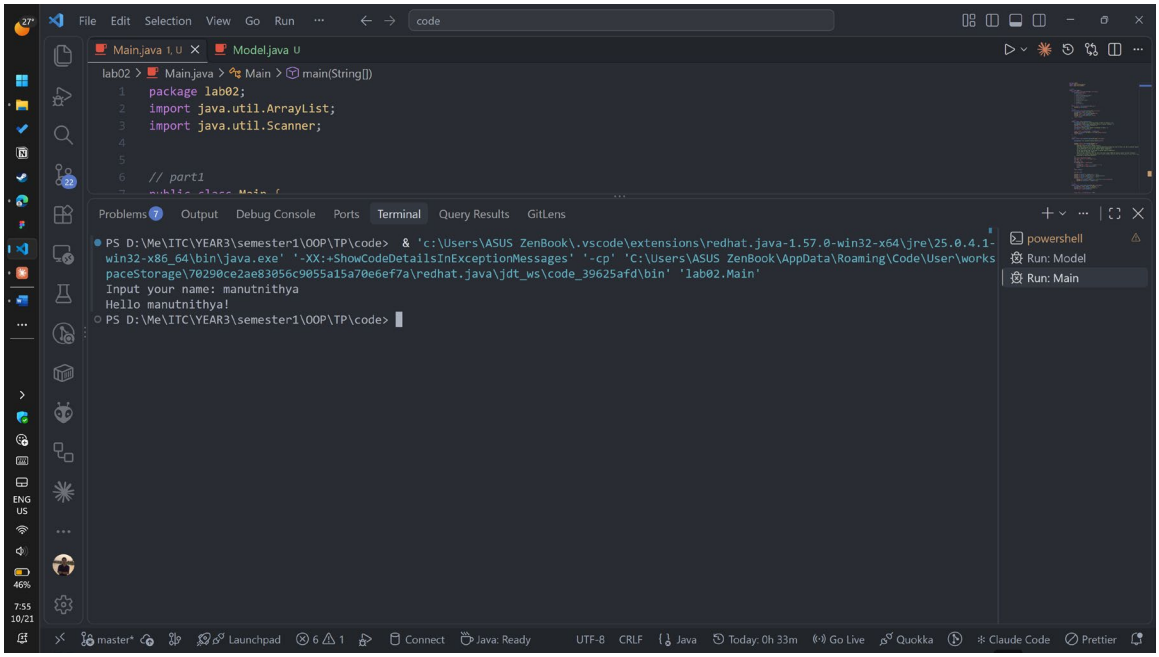
LAB 02 - Basics in Java: Solutions, Source Code and Output

Lab02.1

```
import java.util.Scanner;

public class Ex01_HelloName {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);
        System.out.print("Input your name: ");
        String name = input.nextLine();
        System.out.println("Hello " + name + "!");
        input.close();
    }
}
```

Output (console)



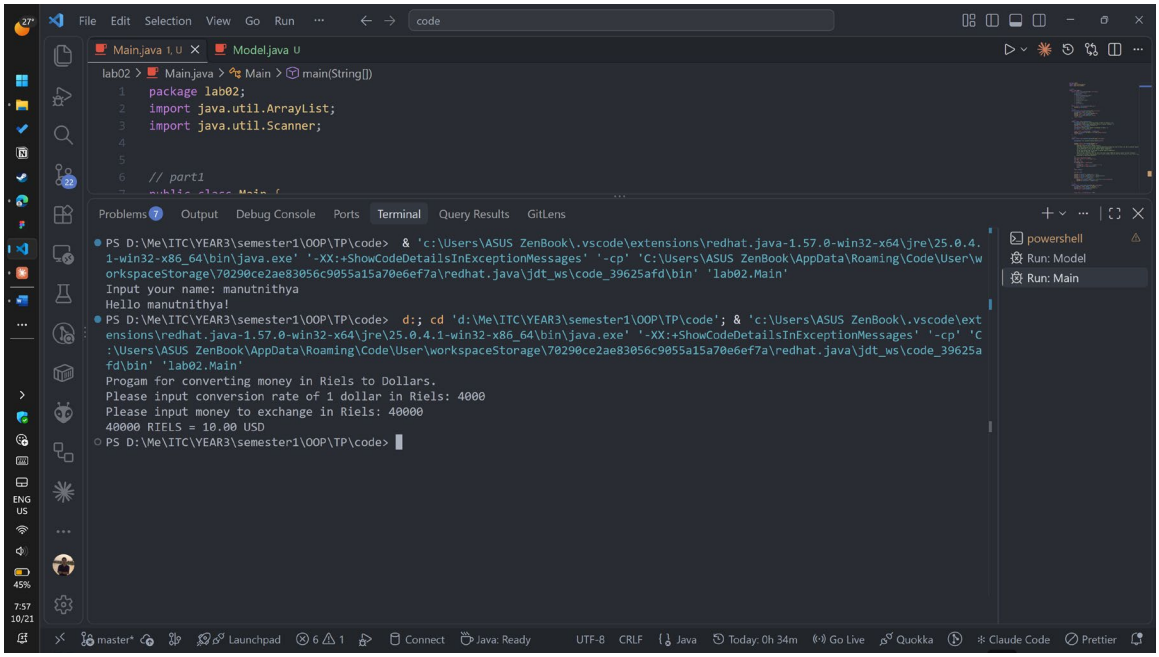
Lab02.2

```
import java.util.Scanner;

public class Ex02_CurrencyConverter {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);
        System.out.println("Program for converting money in Riels to
Dollars.");
        System.out.print("Please input conversion rate of 1 dollar in Riels:
");
        double rate = input.nextDouble();
        System.out.print("Please input money to exchange in Riels: ");
```

```
        double riels = input.nextDouble();
        double usd = riels / rate;
        System.out.printf("%.0f RIELS = %.2f USD%n", riels, usd);
        input.close();
    }
}
```

Output (console)



Lab02.3

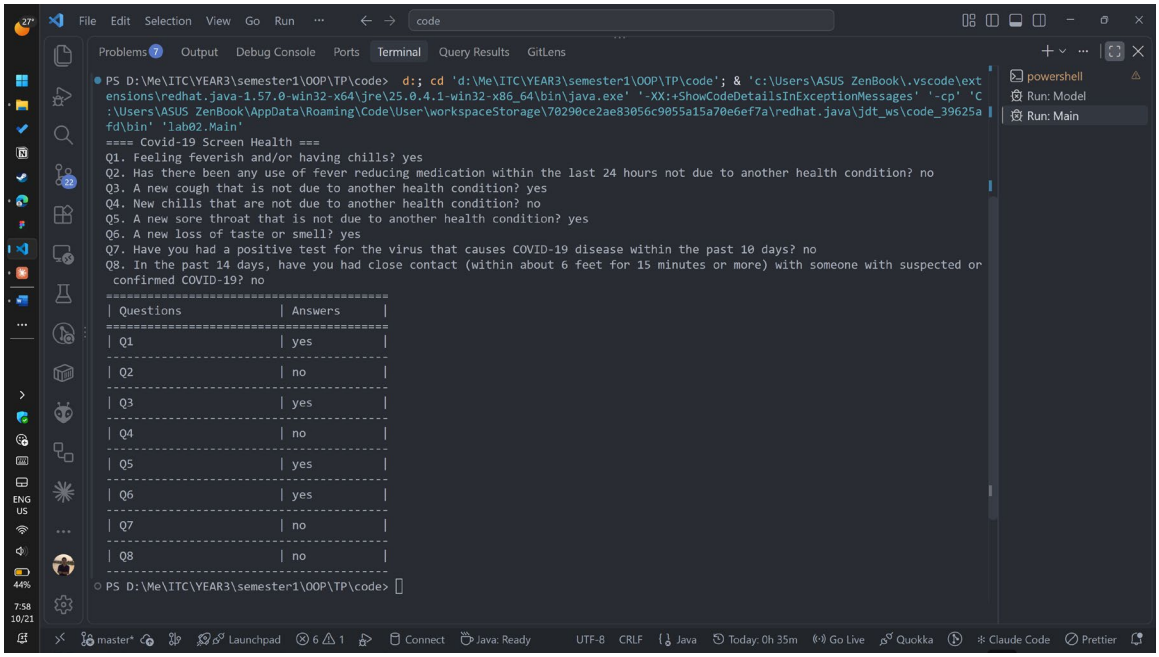
```
import java.util.Scanner;

public class Ex03_HealthScreening {
    public static void main(String[] args) {
        String[] questions = {
            "Feeling feverish and/or having chills?",
            "Has there been any use of fever reducing medication within the
last 24 hours not due to another health condition?",
            "A new cough that is not due to another health condition?",
            "New chills that are not due to another health condition?",
            "A new sore throat that is not due to another health condition?",
            "A new loss of taste or smell?",
            "Have you had a positive test for the virus that causes COVID-19
disease within the past 10 days?",
            "In the past 14 days, have you had close contact (within about 6
feet for 15 minutes or more) with someone with suspected or confirmed COVID-
19?"
        };
        Scanner input = new Scanner(System.in);
        System.out.println("==== Covid-19 Screen Health ===");
```

```
        String[] answers = new String[questions.length];
        for (int i = 0; i < questions.length; i++) {
            System.out.print("Q" + (i + 1) + ". " + questions[i] + " ");
            answers[i] = input.nextLine();
        }
        input.close();

        String line = "=".repeat(41);
        System.out.println(line);
        System.out.printf("| %-10s | %-8s |%n", "Questions", "Answers");
        System.out.println(line);
        for (int i = 0; i < answers.length; i++) {
            System.out.printf("| %-9s | %-8s |%n", "Q" + (i + 1), answers[i]);
            System.out.println("-".repeat(41));
        }
    }
}
```

Output (console)



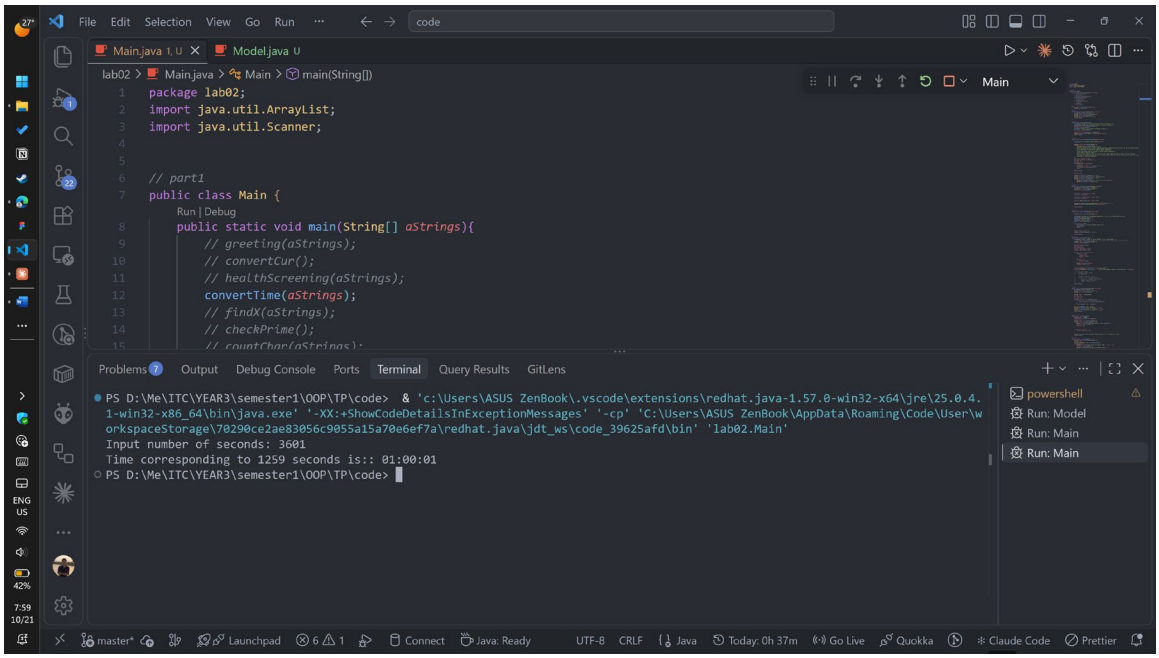
Lab02.4

```
import java.util.Scanner;

public class Ex04_SecondsToTime {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);
        System.out.print("Input number of seconds: ");
        int seconds = input.nextInt();
        int hours = seconds / 3600;
```

```
        int minutes = (seconds % 3600) / 60;
        int secs = seconds % 60;
        System.out.printf("Time corresponding to %d seconds is::
%02d:%02d:%02d%n",
                seconds, hours, minutes, secs);
        input.close();
    }
}
```

Output (console)



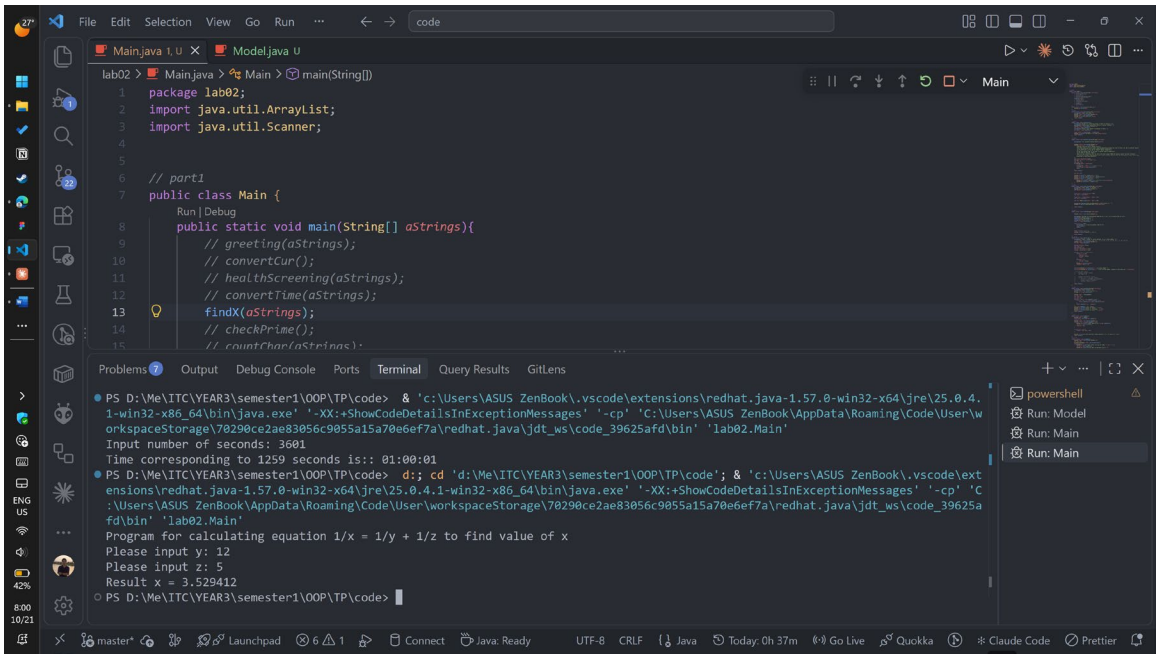
Lab02.5

```
import java.util.Scanner;

public class Ex05_Equation {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);
        System.out.println("Program for calculating equation 1/x = 1/y + 1/z to
find value of x");
        System.out.print("Please input y: ");
        double y = input.nextDouble();
        System.out.print("Please input z: ");
        double z = input.nextDouble();
        if (y == 0 || z == 0 || y + z == 0) {
            System.out.println("y and z must be non-zero and y + z must not be
zero!");
        } else {
            // 1/x = 1/y + 1/z  =>  x = y*z / (y+z)
            double x = (y * z) / (y + z);
            System.out.printf("Result x = %.8f%n", x);
```

```
        }
        input.close();
    }
}
```

Output (console)



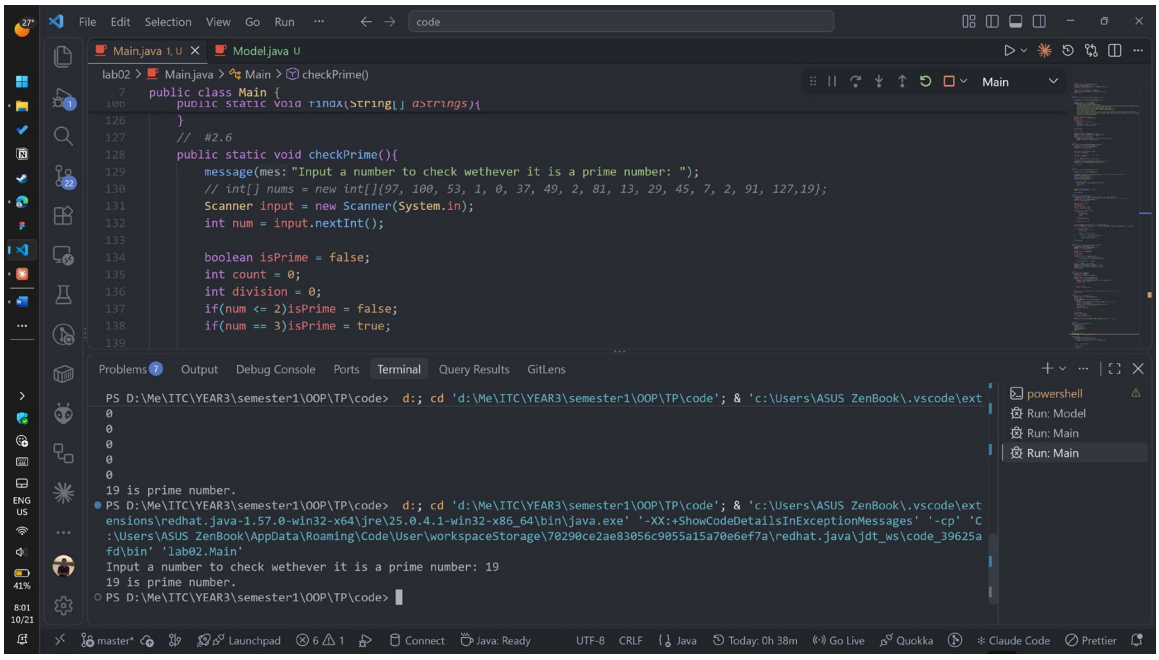
Lab02.6

```
import java.util.Scanner;

public class Ex06_PrimeNumber {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);
        System.out.print("Input a number to check whether it is a prime number:
");
        int num = input.nextInt();
        int divisor = -1;
        for (int i = 2; i * i <= num; i++) {
            if (num % i == 0) {
                divisor = i;
                break;
            }
        }
        if (num < 2) {
            System.out.println(num + " is not a prime number.");
        } else if (divisor == -1) {
            System.out.println(num + " is a prime number.");
        } else {
            System.out.println(num + " is not a prime number, because it is
divisible by " + divisor + ".");
```

```
        }
        input.close();
    }
}
```

Output (console)



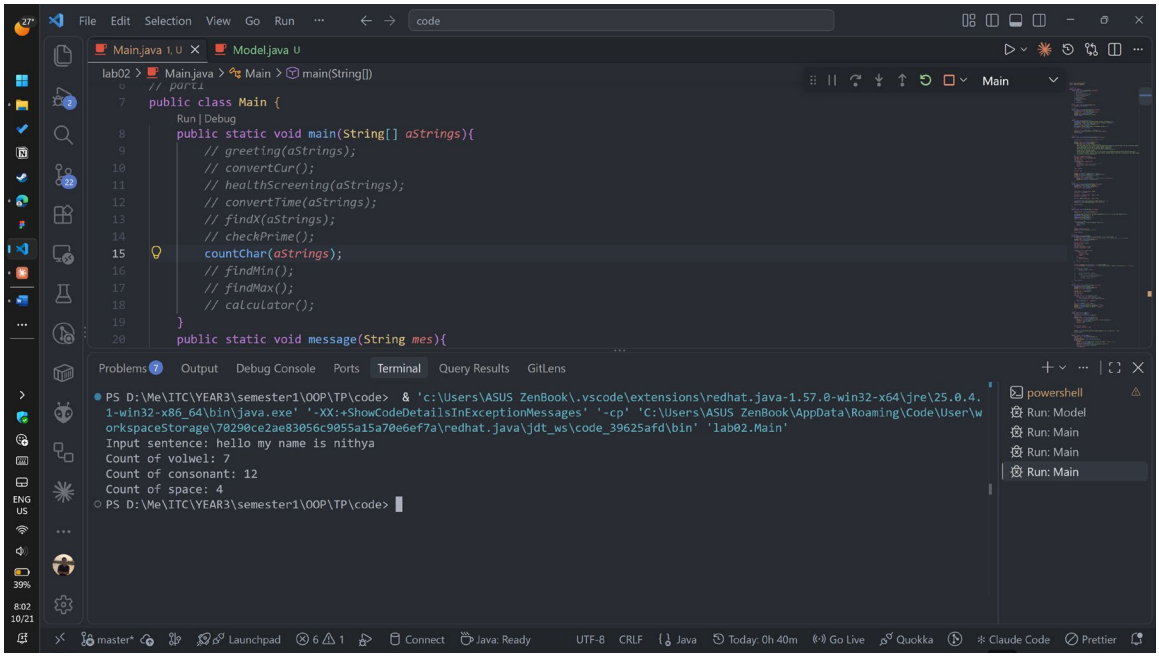
Lab02.7

```
import java.util.Scanner;

public class Ex07_CharacterCounter {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);
        System.out.print("Input a sentence: ");
        String str = input.nextLine().toLowerCase();
        String vowels = "aeiou";
        int vowelCount = 0, consonantCount = 0, spaceCount = 0;
        for (char c : str.toCharArray()) {
            if (c == ' ') {
                spaceCount++;
            } else if (vowels.indexOf(c) >= 0) {
                vowelCount++;
            } else if (Character.isLetter(c)) {
                consonantCount++;
            }
        }
        System.out.println("Count of vowel is: " + vowelCount);
        System.out.println("Count of consonant is: " + consonantCount);
        System.out.println("Count of space is: " + spaceCount);
        input.close();
```

```
    }
}
```

Output (console)

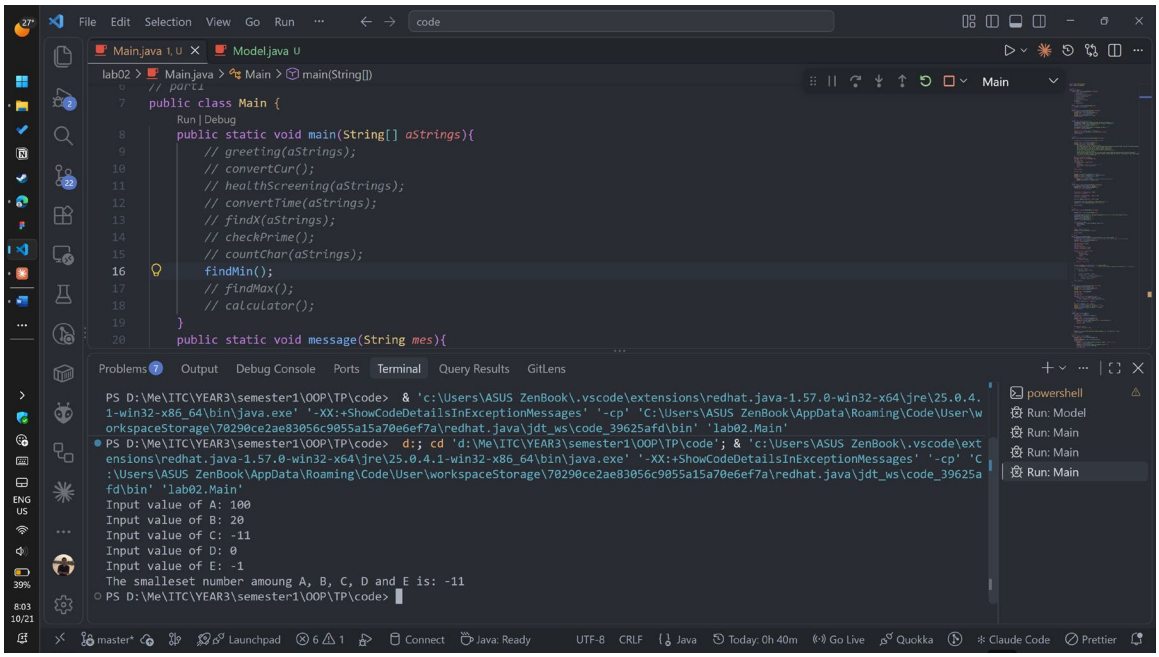


Lab02.8

```
import java.util.Scanner;

public class Ex08_MinimumValue {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);
        char[] names = {'A', 'B', 'C', 'D', 'E'};
        int[] nums = new int[names.length];
        for (int i = 0; i < names.length; i++) {
            System.out.print("Input value of " + names[i] + ": ");
            nums[i] = input.nextInt();
        }
        int min = nums[0];
        for (int num : nums) {
            if (num < min) min = num;
        }
        System.out.println("The smallest number among A, B, C, D, and E is: " +
min);
        input.close();
    }
}
```

Output (console)

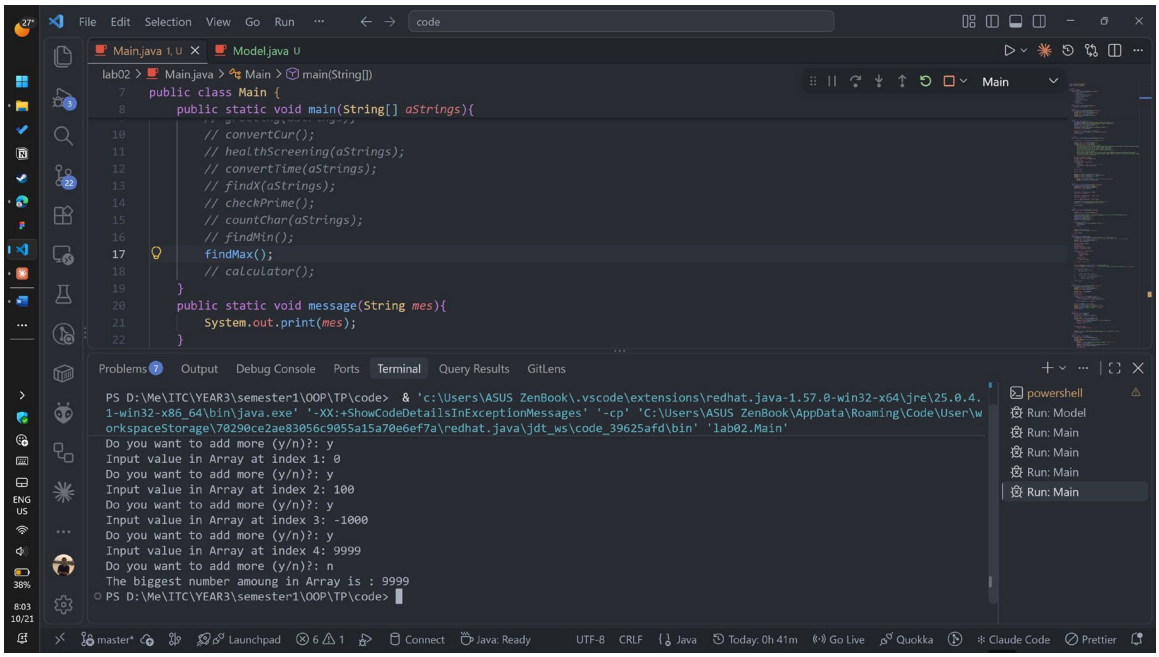


Lab02.9

```
import java.util.ArrayList;
import java.util.Scanner;

public class Ex09_MaximumValue {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);
        ArrayList<Integer> nums = new ArrayList<>();
        int idx = 0;
        while (true) {
            System.out.print("Input value in Array at index " + idx + ": ");
            nums.add(input.nextInt());
            System.out.print("Do you want to add more (y/n)?: ");
            char ans = input.next().charAt(0);
            if (ans != 'y' && ans != 'Y') break;
            idx++;
        }
        int max = nums.get(0);
        for (int num : nums) {
            if (num > max) max = num;
        }
        System.out.println("The biggest number among in Array is: " + max);
        input.close();
    }
}
```

Output (console)



Lab02.10

```
// #2.10
public static void calculator() {
String[] operations = {
    "Addition (+)",
    "Substraction (-)",
    "Multiplication (X)",
    "Division (/)",
    "Quit"
};
String ops = "+-*/";
Scanner input = new Scanner(System.in);

while (true) {
    message("=== Math Menu ===\n");
    for (int i = 0; i < operations.length; i++) {
        System.out.println((i + 1) + " " + operations[i]);
    }

    int op = input.nextInt();

    if (op == 5) {
        break;
    } else if (op < 1 || op > 4) {
        System.out.println("Invalid option!!!");
        continue;
    }

    char sign = ops.charAt(op - 1);

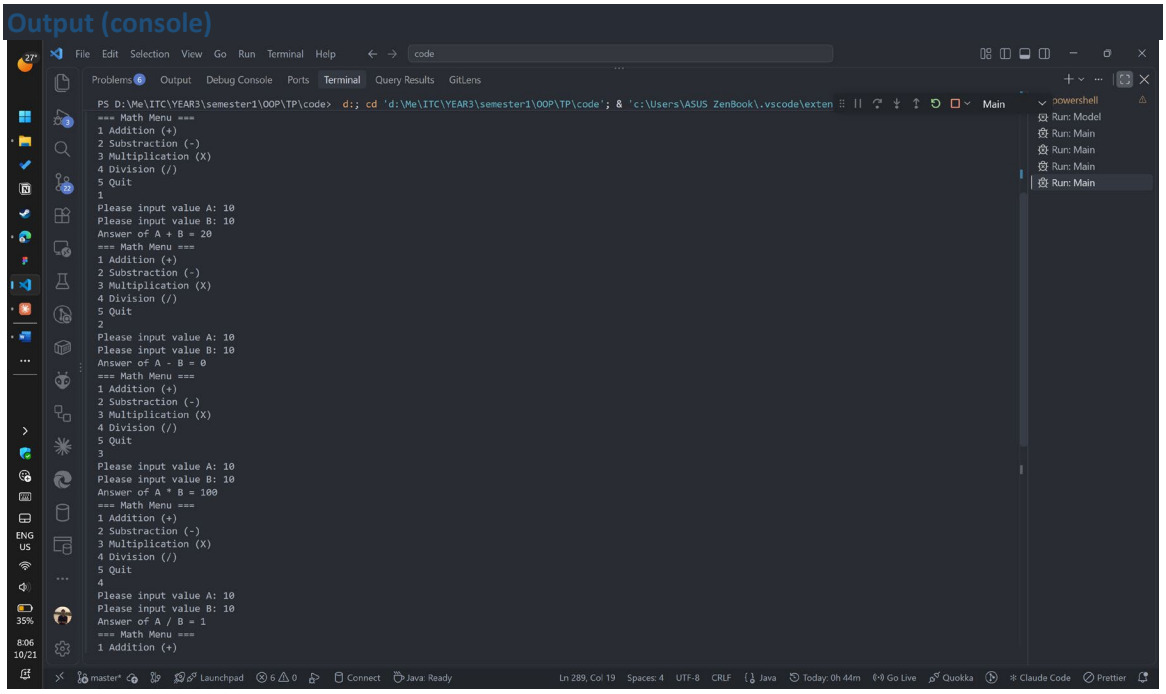
    message("Please input value A: ");
```

```
        int A = input.nextInt();
        message("Please input value B: ");
        int B = input.nextInt();

        int res = 0;
        switch (op) {
            case 1:
                res = A + B;
                break;
            case 2:
                res = A - B;
                break;
            case 3:
                res = A * B;
                break;
            case 4:
                if (B == 0) {
                    System.out.println("Can't divide with 0!!!");
                    continue;
                }
                res = A / B;
                break;
        }

        System.out.printf("Answer of A %c B = %d%n", sign, res);
    }
    input.close();
}
```

Output (console)



Lab02.11

```
package lab02;

// part2
public  class Model{
    // #2.11
    public static void main(String[] aStrings){
        IDCard idc = new IDCard();
        idc.cardNumber = 000001;
        idc.name = "Nithya";
        idc.issuedDate = "12-Dec-2020";
        idc.expireDate = "12-Dec-2030";
        idc.dob = "12-Dec-2000";

        FacebookPost fbp = new FacebookPost();
        fbp.postID = 0001;
        fbp.postUrl = "facebook.come";
        fbp.postDate = "21-Oct-2026";
        fbp.like = 1000000;
        fbp.title = "My familiy trip";
        fbp.desc = "I visited Koh Rong";

        Phone ph = new Phone();
        ph.userID = 1001;
        ph.phoneNum = 12345678;
        ph.userName = "Sean";
        ph.MONs = "Smart";
        ph.registerDate = "2026-10-21";

        IDCard[] idcArr = new IDCard[1];
        idcArr[0] = idc;

        FacebookPost[] fbpArr = new FacebookPost[1];
        fbpArr[0] = fbp;

        Phone[] phArr = new Phone[1];
        phArr[0] = ph;

    }

}
class IDCard{
    int cardNumber;
    String name;
    String issuedDate;
    String expireDate;
    String dob;
}
class FacebookPost{
    int postID;
    String postUrl;
    String postDate;
    int like;
    String title;
```

```
    String desc;
}
class Phone{
    int userID;
    int phoneNum;
    String userName;
    String MONs;
    String registerDate;
}
class Book{
    String title;
    double price;
    String produceDate;
    int page;
    String desc;
}
```